BISHMAY BARIK

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Education

M.Sc. Economics | **Shiv Nadar University, Delhi-NCR** *CGPA: 8.97/10 (3.6/4.0)* | ***Dean's List, 2025*** 2023-2025

- **Subjects:** Macroeconomics (I & II) *(Distinction)*, Econometrics (I & II), Microeconomics (I & II), Economics of Discrimination, Development Economics, Behavioural Economics
- **Dissertation:** *Caste in Stone? Cultural Distance and the Education Game*

B.Sc. Economics (Hons.) | **XIM University, Bhubaneswar** *CGPA: 9.47/10 (3.8/4.0)* | ***Dean's List, 2023*** 2020-2023

- *Merit Scholarship recipient for three consecutive years*
- **Subjects:** Macroeconomics, Microeconomics, Development Economics, International Economics, Game Theory and Behavioural Economics
- **Dissertation:** *Economics of Discrimination: Review of Gary Becker's work and its application on India*

Work In Progress

Consumption Dispersion, Jati Networks, and Risk-Sharing in India Work in Progress

with **Dr. Satyajit Chatterjee**, **Dr. Ashwini Deshpande** & **Dr. Nirupama Kulkarni**

Social networks shape households' ability to insure against income shocks. We show that consumption risk-sharing in India is organized around jati networks, highlighting the continued importance of informal institutions.

Research Experience

Research Associate | **Centre for Advanced Financial Research and Learning (CAFRAL), RBI** July 2025 - Present

Full-time position at Reserve Bank of India, Mumbai, Maharashtra, India

- Worked on two large projects: a typology of the complaints people file with the Banking Ombudsman (with CEPD), and a study of cyber frauds in India (with DoS).
- Built the natural language processing and machine learning behind both projects, sorting over 150,000 complaints written in English, Hindi and Hinglish into policy-relevant categories, with human review to check the results
- Used agentic AI workflows, calling OpenAI and Anthropic models over their APIs to run this at scale, masking all personal information before any data left the machine, and shared the findings as interactive dashboards

Research Assistant | **QuantEcon** May 2025 - Dec 2025

Principal Investigator: Dr. John Stachurski

- Designing open-access undergraduate economics course using Pyodides/WebAssembly for real-time simulation, teaching scientific computing for inequality, resource scarcity, and climate change
- Enhancing QuantEcon's intermediate Python lectures by rewriting selected series with JAX, improving computational efficiency, numerical accuracy, and scalability

Data Science Intern | **Development Data Lab** Mar - Sept 2025

Principal Investigator: Dr. Paul Novosad

- Cleaning and coding Annual Status of Education Reports Data (2007-2022) for incorporation into The Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG) Dataset by Development Data Lab

Research Assistant | **Ashoka University** Apr - May 2025

Principal Investigator: Dr. Bhaskar Dutta

- Replicating methodologies from Journal of Development Economics paper using Annual Status of Education Reports Data at state level

Research Assistant | "Punishment and Crime: Impact of Death Penalty on Local Crime Rates in India" Dec 2024 - Jan 2025

Authors: Dr. Ashwini Deshpande & **Dr. Anisha Sharma**

Research Intern | **CAFRAL, RBI** 2024

Principal Investigators: Dr. Gautham Udupa & **Dr. Vidhya Soundararajan**

- Converting Stata codes to MATLAB equivalents; reviewing RBI circulars on Priority Sector Lending; reviewing literature on developmental factors affecting stock market participation

Public Contributions

Historical Population Census 1931: *Jati* Counts across Granular Regions 2026

Tools: Python (OpenCV, `pdfplumber`, NumPy, `pandas`), Anthropic Batch API, Poppler, Stata

- Transcribes caste (*jati*) population counts from 26 scanned provincial and princely-state volumes of the 1931 Census (Imperial Table XVII) into a labelled Stata panel: 69,305 rows across 49 provinces/states, 1,849 caste names and 1,200 districts, towns and tehsils, with generated codebook and validation report
- Computer-vision page segmentation before any model call: native-DPI rendering, Tesseract OSD deskewing, and OpenCV ink-projection profiles that place every tile cut in whitespace on the table's own column-group boundaries, keeping each caste's Persons/Males/Females intact and replicating the row-label stub into every tile

- Two independent vision passes per tile via the Batch API (2,174 readings), cached on a pixel-level MD5 digest so tiler improvements re-read only changed tiles; validated by column-alignment, sex-sum and geographic-hierarchy checks, with rival readings retained rather than discarded

lipi-swap: Devanagari to Roman Transliteration Utility 2025

Tools: Python (`requests`, `BeautifulSoup`, `pandas`, `unidecode`)

- Python utility for converting Devanagari script (Hindi) into Roman English characters, solving practical challenges in handling non-Latin text when working with Indian administrative datasets
- Integrates transliteration engine into web scraping workflow for hierarchical Indian administrative data (states, districts, blocks, panchayats); ensures datasets remain readable and compatible across platforms
- Eliminates encoding issues in CSV exports by converting geographic entities labeled in Hindi script (e.g., "पंचायत " → "panchayat")

State-wise *Jati* Distributions Dashboard 2025

Tools: Python (Streamlit, pandas), DuckDB, Apache Parquet, AWS S3

- Interactive Streamlit dashboard visualizing caste (*jati*) distribution across India using CMIE-CPHS (2014-2024); dynamic filters by state/district/year/month, highlights dominant *jatis*
- Direct cloud integration: queries remote Parquet files from AWS S3 without local downloads; highly optimized SQL queries via DuckDB for efficient data retrieval over HTTPS
- Performance optimization: `@st.cache_data` caching for metadata loading and database queries; columnar storage format enables fast aggregation without loading full dataset into memory

Dissertation and Research Projects

Caste in Stone? Cultural Distance and the Education Game (*Master's Thesis*) 2024-25

Supervisors: Dr. Partha Chatterjee, **Dr. Pubali Chakraborty**

- Developed three-period macroeconomic model analyzing intergenerational human capital/income disparities across caste/religious groups using CPHS and Samaj Survey data
- Quantified cultural distance's role in perpetuating intergroup disparities, linking cultural norms variation to human capital accumulation and income outcomes

Shedding Light on Consumption: Night-Time Luminosity as Predictor of Consumption Expenditure in India 2025

Tools: Python (Streamlit, Plotly, TensorFlow, Pillow)

- Exploring night-time light intensity correlation with consumption expenditure using SECC/DMSP Nightlights data from SHRUG
- Neural network with TensorFlow/Keras for prediction; SHAP analysis for model interpretability

Tracing Disparities: Educational Spending Before/After Covid-19 Among Social Groups in India 2024

Supervisor: Dr. Gitanjali Sen | *Co-author: Manya Pahwa* | *Tools: Python, STATA*

- Difference-in-Differences model estimating pre/post-Covid educational expenditure trends across religion/caste groups using CMIE-CPHS
- Explored reasons for differential social group impacts during pandemic

Does Overconfidence Lead to Procrastination?: A Cognitive Reflection Test Experiment 2024

Supervisor: Dr. Punarjit Roychowdhury

- Bridge between overconfidence and procrastination using Cognitive Reflection Test to identify overconfident individuals and study their procrastination behavior

Environmental Kuznets Curve (EKC) Analysis: Case Study on Ganga and Yamuna 2024

Supervisor: Dr. Ram Ranjan | *Co-author: Manya Pahwa* | *Tools: STATA, R, Excel*

- Econometric model testing EKC in Ganga/Yamuna; analyzed pollution patterns and identified causes

Economics of Discrimination: Review of Gary Becker's work and its application on India (*B.Sc. Dissertation*) 2022-23

Institution: XIM University, Bhubaneswar | **Supervisors: Dr. Bharatee Bhusana Dash**, **Dr. Pintu Parui**

Media and Policy

The FRED Blog (FRB St. Louis) | Economic policy uncertainty and aggregate economic activity in India May 2026

with **Dr. B. Ravikumar**

The Policy Edge | India's Digital Divide Is Now a Capability Gap Mar 2026

with **Dr. Shreya Bhattacharya**

Teaching and Training

AI Workshops and Training | CAFRAL and Reserve Bank of India 2026

- **3 Aug 2026:** Session for **CAFRAL Learning and Administration**, introducing AI tools and workflow techniques

- **Mar 2026:** Five sessions for the **CAFRAL Research Team** on agentic coding with Claude Code: skills and sub-agents, compound engineering workflows, running Stata and Python through AI agents, and sandboxing agents in Docker. Followed by a hands-on session writing a paper end to end using these tools

Graduate Teaching Assistant | **Shiv Nadar University, India** 2023-2024

- **Semester IV (Postgraduate):** ECO512: Macroeconomics II | **Dr. Kurt Horner**, **Dr. Trishita Ray Barman**
- **Semester III:** ECO302: Intermediate Macroeconomics | **Dr. Aditi Singh**, **Dr. Trishita Ray Barman**
- **Semester II:** ECO461: International Economics | **Dr. Trishita Ray Barman**
- **Semester I:** MAT101: Calculus I | **Dr. Niteesh Sahni**

Software

Python | STATA | LaTeX | Markdown | Jupyter Notebook | Jupyterlab Myst

Technical Tools & Workflow

Machine Learning & NLP: PyTorch, Hugging Face Transformers (fine-tuned MuRIL/BERT classifiers for Hindi and Hinglish text), sentence-transformers embeddings, scikit-learn, XGBoost; clustering, active learning, and probability calibration
LLMs & Agentic Workflows: OpenAI and Anthropic APIs (Batch API, prompt caching, JSON-schema-constrained decoding, teacher-student distillation); vision-model document extraction from scanned archival sources; local open-weight inference via Ollama; PII masking pipelines; agentic development with Claude Code
Computer Vision & Document Processing: OpenCV (adaptive thresholding, ink-projection profiles for column/row segmentation, morphological rule detection); `pdfplumber` and Poppler (`pdftoppm`) for native-resolution page rendering; Tesseract OSD deskewing; automated table tiling and reconstruction
Data & Visualisation: Apache Parquet, DuckDB, Dash, Plotly, Streamlit
Workflow & Development: Git/GitHub (CLI-based version control and pipeline management), Bash scripting, GNU Make (project pipeline automation); reproducible research pipelines with content-addressed caching and validation gates; Stata `.dta` outputs with variable labels and generated codebooks

References

Available on request.